Shulba Sutras (approx. 800 BCE to 500 BCE) are among the world's oldest mathematical texts, primarily detailing the geometry required for constructing complex Vedic altars. Later, **Brahmagupta** (7th Century CE) revolutionized algebra and geometry.

2.1 Ancient Knowledge from the Shulba Sutras

2.1.1 Construction of a Square

Goal: Create a square that has the same area as a given circle.

Source: Baudhayana Sulba Sutra, 1.59

This is one of the classic problems of antiquity. Baudhayana gives an approximate geometric solution.

The Sanskrit Sutra

मण्डलं चतुरस्रं चिकीर्षन्विष्कम्भमष्टौ भागान्कृत्वा भागमेकोनत्रिंशधा

विभज्याष्टाविंशति भागानुद्धरेत्भागस्य च षष्ठमष्टम भागोनम् ॥

Maṇḍalaṃ caturasraṃ cikīrṣan viṣkambham aṣṭau bhāgān kṛtvā bhāgam ekona-triṃśadhā vibhajya aṣṭāviṃśati bhāgān uddharet bhāgasya ca ṣaṣṭham aṣṭama-bhāgonam ||

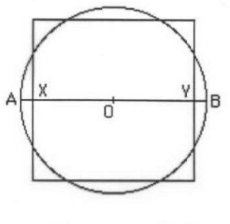
Meaning & Method

"If you wish to turn a circle into a square (of equal area):"

1. Divide the **Diameter** of the circle into **8 equal parts**.
2. Take one of those 8 parts and divide it further into **29 equal sub-parts**.
3. Reduce the diameter by **28 of those 29 sub-parts** (so you are left with 1/29 of that 1/8th part) and a tiny correction factor (one-sixth of that part minus one-eighth of that).

Simplified Rule:

Effectively, the side of the square (a) is related to the diameter (d) by:



The Sulbasutras 13/15 method of "circling the square"

$$a = d\left(1 - \frac{1}{8} + \frac{1}{8\cdot 29} - 1/(8*29*6) + 1/(8*29*6*8)\right)$$

```python
def solve_baudhayana_squaring_circle(diameter):
    """
    Calculates the side of a square ('a') that has approximately the same area
    as a circle with diameter ('d'), using the Baudhayana Sulba Sutra formula.
    """
    print(f"--- Squaring the Circle (Baudhayana Sulba Sutra) ---")
    print(f"Input Diameter (d): {diameter}")

    # 1. Define the terms of the fraction series exactly as written in the Sutra
    # Formula: 1 - 1/8 + 1/(8*29) - 1/(8*29*6) + 1/(8*29*6*8)

    term_1 = 1.0
    term_2 = 1 / 8
    term_3 = 1 / (8 * 29)
    term_4 = 1 / (8 * 29 * 6)
    term_5 = 1 / (8 * 29 * 6 * 8)

    # 2. Calculate the scaling factor
    # Note: The signs alternate: +, -, +, -, +
    factor = term_1 - term_2 + term_3 - term_4 + term_5

    # 3. Calculate the side of the square 'a'
    side_a = diameter * factor

    print(f"\n[Calculation Steps]")
    print(f"1. Base Unit (1)              : {term_1}")
    print(f"2. Subtract 1/8               : -{term_2:.6f}")
    print(f"3. Add 1/(8*29)               : +{term_3:.6f}")
    print(f"4. Subtract 1/(8*29*6)        : -{term_4:.6f}")
```

```
    print(f"5. Add 1/(8*29*6*8)          : +{term_5:.6f}")
    print(f"------------------------------------------")
    print(f"Total Scaling Factor         : {factor:.8f}")

    print(f"\n[Results]")
    print(f"Calculated Side of Square (a): {side_a:.5f}")

    # --- Verification & Accuracy Check ---
    square_area = side_a ** 2
    circle_area = math.pi * (diameter / 2) ** 2

    # Implicit Pi value in this Sutra:
    # If Area = Pi * (d/2)^2 and Area approx equals a^2
    # Then Pi_approx = 4 * (a/d)^2
    sutra_pi = 4 * (factor ** 2)

    print(f"\n[Accuracy Check]")
    print(f"Area of Square (a^2)         : {square_area:.5f}")
    print(f"Area of Circle (Modern Pi)   : {circle_area:.5f}")
    print(f"Difference                   : {abs(square_area - circle_area):.5f}")
    print(f"\nImplied Value of Pi in Sutra : {sutra_pi:.5f}")
    print(f"Modern Value of Pi           : {math.pi:.5f}")

# --- Example Usage ---
# You can change the diameter below to any number (e.g., 10, 50, 100)
solve_baudhayana_squaring_circle(diameter=100)
```

This gives a value of pi approx 3.088, which was sufficient for brick construction.

2.1.2 The Pythagorean Theorem (Shulba Sutra 1.48/1.2)

Long before Pythagoras, the *BaudhayanaShulba Sutra* recorded this geometric principle.

Sutra:

दीर्घचतुरश्रस्याक्ष्णया रज्जुः पार्श्वमानी तिर्यङ्गानी च।

यत्पृथग्भूते कुरुतस्तदुभयं करोति ॥

dīrghasyākṣaṇayā rajju ḥpārśvamānī tiryaḍmānī cha

yatpṛthagbhūte kurutastadubhayāṅ karoti ||

English Meaning:

"The rope stretched along the diagonal (akṣaṇayā) of a rectangle produces an area which the vertical and horizontal sides make together."

Example:

1. A rectangle with sides 3 and 4.
2. The square of the diagonal (5^2 = 25) equals the sum of the squares of the sides (3^2 + 4^2 = 9 + 16 = 25).

Python Code:

Python
```
deffind_diagonal(a, b):
return (a**2 + b**2)**0.5

# Example: Sides 3 and 4
print(f"The diagonal of a 3x4 rectangle is: {find_diagonal(3, 4)}")
```

2.1.3 Area of Circle

The Shulba Sutras used a process called "Squaring the Circle" and "Circling the Square" to find approximate areas.

Sutra:

वृत्तक्षेत्रे परिधिगुणितव्यासपादः फलं तत्

(area of circle)

Area= Circumference x (Diameter/4)

क्षुण्णं वेदैरुपरि परितः कन्दुकस्येव जालम्।

(surface area of a Sphere)

Area=4x Area of Circle=4 x pi x r x r

गोलस्यैवं तदपि च फलं पृष्ठजं व्यासनिघ्नं

षड्भिर्भक्तं भवति नियतं गोलगर्भे घनाख्यम्॥

Volume=(surface area x Diameter)/6

Example:

1. If a square has a side of 10 units, the area is 100.
2. The Vedic mathematicians derived a circle radius that would approximate this same area.

Python Code:

Python
```
defcircle_area_vedic(radius):
    # Using an approximate Vedic value for pi (~3.088)
pi_vedic = 3.088
returnpi_vedic * (radius**2)

print(f"Vedic Circle Area (r=5): {circle_area_vedic(5)}")
```

2.1.4 Area of Triangle

The texts describe the area of a triangle as half the area of the rectangle that encloses it.

The Logic:

$$Area = \frac{1}{2} \times Base \times Height$$

Example:

1. A triangle with base 10 and height 5.
2. Area = 0.5 \times 10 \times 5 = 25.

Python Code:

Python
```
deftriangle_area(base, height):
return 0.5 * base * height

print(f"Triangle Area (b=10, h=5): {triangle_area(10, 5)}")
```

2.2 Ancient Knowledge by Brahmagupta

2.2.1 Area of Cyclic Quadrilateral

Brahmagupta provided the formula for the area of a quadrilateral whose vertices lie on a circle.

Sutra (Brahmasphutasiddhanta, Verse 12.21):

स्थूलफलंत्रिचतुर्भुजबाहूप्रतिबाहुघातार्धम् ।

अर्धयुतभुजघातवधःसपदःपदंसूक्ष्मम् ॥

- स्थूल → स्थूल / मोटा / अनुमानित
- फलम् → परिणाम (area)
- त्रि-चतुर्भुज → त्रिभुज और चतुर्भुज
- बाहु → भुजा (side)
- प्रति-बाहु → आमने-सामने की भुजा
- घात → गुणन
- अर्धम् → आधा
- अर्ध-युत → आधा जोड़कर
- भुज-घात → भुजाओं का गुणन
- वधः → घटाने पर
- स-पदः → साथ में पद (correction term)
- पदम् → पद / चरण
- सूक्ष्मम् → सूक्ष्म / सटीक

स्थूल फलम् त्रि चतुर्भुज बाहु उप्रति बाहु घात अर्धम् ।

अर्ध युत भुज घात वधः स पदः पदम् सूक्ष्मम् ॥

English Meaning:

"The square root of the product of the half-sum of the sides diminished by each side is the exact area."

Formula:

If sides are a, b, c, d and semi-perimeter $s = \frac{(a+b+c+d)}{2}$

Area = $\sqrt{((\boldsymbol{s}-\boldsymbol{a})(\boldsymbol{s}-\boldsymbol{b})(\boldsymbol{s}-\boldsymbol{c})(\boldsymbol{s}-\boldsymbol{d}))}$

Example:

1. Sides of a cyclic quadrilateral: a=7, b=15, c=20, d=24.
2. Calculate s and then the Area.

Python Code:

```
Def brahmagupta_area(a, b, c, d):
    s = (a + b + c + d) / 2
area = ((s-a) * (s-b) * (s-c) * (s-d))**0.5
return area

# Example sides
print(f"Area of Cyclic Quadrilateral: {brahmagupta_area(7, 15, 20, 24)}")
```